KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5061 | 5 Credits: 4 | Program core | Source-grounded

Textile Testing

 To impart the knowledge of statistical techniques and humidity in textile testing and

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5061 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5061.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Textile Technology
Course code	5061
Course title	Textile Testing
Semester	5 Credits: 4
Course category	Program core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart the knowledge of statistical techniques and humidity in textile testing and
- To understand the principles and methods for testing of fibres, yarn and fabrics.

Course outcomes

CO	Official outcome	Study level
CO1	and influence of moisture and Humidity on textile 15 Applying materials. Explain the procedure for testing Fiber length,	See official syllabus
CO2	strength, fineness, maturity and trash content in 15 Understanding cotton. Recognize the Yarn testing methods for	See official syllabus
CO3	determining count, strength, twist, evenness and 14 Understanding hairiness of yarn. Discuss the Fabric testing methods for determining	See official syllabus
CO4	14 Understanding various fabric properties.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	moisture and Humidity on textile materials.	4	As specified
Module 2	maturity and trash content in cotton.	4	As specified
Module 3	twist, evenness and hairiness of yarn.	4	As specified
Module 4	properties.	4	As specified

MODULE 1

moisture and Humidity on textile materials.

This module is presented from the official syllabus scope for Textile Testing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: moisture and Humidity on textile materials.
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

Understand the objectives of Textile Testing 2 Remembering Understand the elements of statistics and its

Understand the objectives of Textile Testing 2 Remembering Understand the elements of statistics and its is an official topic in Module 1 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the objectives of Textile Testing 2 Remembering Understand the elements of statistics and its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the objectives of textile testing 2 remembering understand the elements of statistics and its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand the objectives of Textile Testing 2 Remembering Understand the elements of statistics and its means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Understand the objectives of Textile Testing 2 Remembering Understand the elements of statistics and its □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

6 Applying application in textile testing. Understand the influence of moisture and

6 Applying application in textile testing. Understand the influence of moisture and is an official topic in Module 1 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying application in textile testing. Understand the influence of moisture and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying application in textile testing. understand the influence of moisture and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying application in textile testing. Understand the influence of moisture and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 6 Applying application in textile testing. Understand the influence of moisture and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding Humidity on textile materials. Understand the measurement of Humidity, Understanding

3 Understanding Humidity on textile materials. Understand the measurement of Humidity, Understanding is an official topic in Module 1 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Humidity on textile materials. Understand the measurement of Humidity, Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding humidity on textile materials. understand the measurement of humidity, understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Humidity on textile materials. Understand the measurement of Humidity, Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding Humidity on textile materials. Understand the measurement of Humidity, Understanding □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Moisture content and regain Objectives of Textile Testing - Define statistics - Selection of Sample for testing - population and sample in statistics - frequency distribution - Graphical representation of frequency distribution - Histogram, frequency polygon and frequency curve - Measures of central Tendency - mean, median and mode - Measures of Dispersion - Range, standard deviation, CV%, Variance . Estimation of Measures of central Tendency & Dispersion. Humidity - importance of humidity in textile testing - expressions of humidity - Absolute humidity and Relative humidity - Measurement of Humidity of the atmosphere by wet and dry bulb and sling hygrometer - standard testing atmosphere - Estimation of Moisture content and regain of textile materials by conditioning oven and electronic moisture meter - Standard moisture regain of important textile fibers - Estimation of standard regain of blends P/C, P/W, P/V - Effect of moisture regain on fibre properties. Explain the procedure for testing Fiber length, strength, fineness,

4 Moisture content and regain Objectives of Textile Testing - Define statistics - Selection of Sample for testing - population and sample in statistics - frequency distribution - Graphical representation of frequency distribution - Histogram, frequency polygon and frequency curve - Measures of central Tendency - mean, median and mode - Measures of Dispersion - Range, standard deviation, CV%, Variance . Estimation of Measures of central Tendency & Dispersion. Humidity - importance of humidity in textile testing - expressions of humidity - Absolute humidity and Relative humidity - Measurement of Humidity of the atmosphere by wet and dry bulb and sling hygrometer - standard testing atmosphere - Estimation of Moisture content and regain of textile materials by conditioning oven and electronic moisture meter - Standard moisture regain of important textile fibers - Estimation of standard regain of blends P/C, P/W, P/V - Effect of moisture regain on fibre properties. Explain the procedure for testing Fiber length, strength, fineness, is an official topic in Module 1 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Moisture content and regain Objectives of Textile Testing - Define statistics - Selection of Sample for testing - population and sample in statistics - frequency distribution - Graphical representation of frequency distribution - Histogram, frequency polygon and frequency curve - Measures of central Tendency - mean, median and mode - Measures of Dispersion - Range, standard deviation, CV%, Variance . Estimation of Measures of central Tendency & Dispersion. Humidity - importance of humidity in textile testing - expressions of humidity - Absolute humidity and Relative humidity - Measurement of Humidity of the atmosphere by wet and dry bulb and sling hygrometer - standard testing atmosphere - Estimation of Moisture content and regain of textile materials by conditioning oven and electronic moisture meter - Standard moisture regain of important textile fibers - Estimation of standard regain of blends P/C, P/W, P/V - Effect of moisture regain on fibre properties. Explain the procedure for testing Fiber length, strength, fineness, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 moisture content and regain objectives of textile testing - define statistics - selection of sample for testing - population and sample in statistics - frequency distribution - graphical representation of frequency distribution - histogram, frequency polygon and frequency curve - measures of central tendency - mean, median and mode - measures of dispersion - range, standard deviation, cv%, variance . estimation of measures of central tendency & dispersion. humidity - importance of humidity in textile testing - expressions of humidity - absolute humidity and relative humidity - measurement of humidity of the atmosphere by wet and dry bulb and sling hygrometer - standard testing atmosphere - estimation of moisture content and regain of textile materials by conditioning oven and electronic moisture meter - standard moisture regain of important textile fibers - estimation of standard regain of blends p/c, p/w, p/v - effect of moisture regain on fibre properties. explain the procedure for testing fiber length, strength, fineness, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Moisture content and regain Objectives of Textile Testing - Define statistics - Selection of Sample for testing - population and sample in statistics - frequency distribution - Graphical representation of frequency distribution - Histogram, frequency polygon and frequency curve - Measures of central Tendency - mean, median and mode - Measures of Dispersion - Range, standard deviation, CV%, Variance . Estimation of Measures of central Tendency & Dispersion. Humidity - importance of humidity in textile testing - expressions of humidity - Absolute humidity

Aspect	What to prepare
	and Relative humidity - Measurement of Humidity of the atmosphere by wet and dry bulb and sling hygrometer - standard testing atmosphere - Estimation of Moisture content and regain of textile materials by conditioning oven and electronic moisture meter - Standard moisture regain of important textile fibers - Estimation of standard regain of blends P/C, P/W, P/V - Effect of moisture regain on fibre properties. Explain the procedure for testing Fiber length, strength, fineness, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Moisture content and regain Objectives of Textile Testing - Define statistics - Selection of Sample for testing - population and sample in statistics - frequency distribution - Graphical representation of frequency distribution - Histogram, frequency polygon and frequency curve - Measures of central Tendency - mean, median and mode - Measures of Dispersion - Range, standard deviation, CV%, Variance . Estimation of Measures of central Tendency & Dispersion. Humidity - importance of humidity in textile testing - expressions of humidity - Absolute humidity and Relative humidity - Measurement of Humidity of the atmosphere by wet and dry bulb and sling hygrometer - standard testing atmosphere - Estimation of Moisture content and regain of textile materials by conditioning oven and electronic moisture meter - Standard moisture regain of important textile fibers - Estimation of standard regain of blends P/C, P/W, P/V - Effect of moisture regain on fibre properties. Explain the procedure for testing Fiber length, strength, fineness, □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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moisture and Humidity on textile materials.
  M1.01      Understand the objectives of Textile Testing        2
Remembering
             Understand the elements of statistics and its
  M1.02                                                          6
Applying
             application in textile testing.
             Understand the influence of moisture and
  M1.03                                                          3
Understanding
             Humidity on textile materials.
             Understand the measurement of Humidity,
Understanding
  M1.04                                                          4
             Moisture content and regain
Contents:
Objectives of Textile Testing - Define statistics - Selection of Sample for
testing -
population and sample in statistics - frequency distribution - Graphical
representation of
frequency distribution - Histogram, frequency polygon and frequency curve -
Measures of
central Tendency - mean, median and mode - Measures of Dispersion - Range,
standard
deviation, CV%, Variance . Estimation of Measures of central Tendency &
Dispersion.
Humidity - importance of humidity in textile testing - expressions of humidity -
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

maturity and trash content in cotton.

This module is presented from the official syllabus scope for Textile Testing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: maturity and trash content in cotton.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Understanding fineness and its determination methods Understand the importance and estimation of

4 Understanding fineness and its determination methods Understand the importance and estimation of is an official topic in Module 2 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding fineness and its determination methods Understand the importance and estimation of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding fineness and its determination methods understand the importance and estimation of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding fineness and its determination methods Understand the importance and estimation of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding fineness and its determination methods Understand the importance and estimation of □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

fibre strength and maturity. Understand the method of estimation of trash

fibre strength and maturity. Understand the method of estimation of trash is an official topic in Module 2 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify fibre strength and maturity. Understand the method of estimation of trash using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, fibre strength and maturity. understand the method of estimation of trash should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What fibre strength and maturity. Understand the method of estimation of trash means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

> **Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ fibre strength and maturity. Understand the method of estimation of trash □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding content in cotton by Shirley trash analyzer.

4 Understanding content in cotton by Shirley trash analyzer. is an official topic in Module 2 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding content in cotton by Shirley trash analyzer. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding content in cotton by shirley trash analyzer. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding content in cotton by Shirley trash analyzer. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 4 Understanding content in cotton by Shirley trash analyzer. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Understand advanced fibre testing systems 2 Understanding List the important properties of fibre - Importance of fiber length - Estimation of fiber length parameters using Baer sorter - Span length, 2.5% and 50% span length, uniformity ratio - Estimation of fiber length using Digital fibro graph. Importance of fiber fineness - Airflow principle - Determination of fiber fineness by ATIRA fiber fineness tester. Importance of fiber strength - Strength terminologies-Load, Breaking load, Stress, Mass stress, Tenacity, Strain, Young's modulus, Elongation, Breaking extension - Estimation of bundle fiber strength by Stelometer (CRL Principle). Importance of fiber maturity - List the different methods of determination of fiber maturity - Estimation of fiber maturity by Caustic soda swelling method. Cleaning efficiency - Lint - Trash - invisible loss - Estimation of trash content in cotton by Shirley trash analyzer. Features and working of Advanced Fiber Information System [AFIS] and High Volume Instrument [HVI]. Recognize the Yarn testing methods for determining count, strength,

Understand advanced fibre testing systems 2 Understanding List the important properties of fibre - Importance of fiber length - Estimation of fiber length parameters using Baer sorter - Span length, 2.5% and 50% span length, uniformity ratio - Estimation of fiber length using Digital fibro graph. Importance of fiber fineness - Airflow principle - Determination of fiber fineness by ATIRA fiber fineness tester. Importance of fiber strength - Strength terminologies-Load, Breaking load, Stress, Mass stress, Tenacity, Strain, Young's modulus, Elongation, Breaking extension - Estimation of bundle fiber strength by Stelometer (CRL Principle). Importance of fiber maturity - List the different methods of determination of fiber maturity - Estimation of fiber maturity by Caustic soda swelling method. Cleaning efficiency - Lint - Trash - invisible loss - Estimation of trash content in cotton by Shirley trash analyzer. Features and working of Advanced Fiber Information System [AFIS] and High Volume Instrument [HVI]. Recognize the Yarn testing methods for determining count, strength, is an official topic in Module 2 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand advanced fibre testing systems 2 Understanding List the important properties of fibre - Importance of fiber length - Estimation of fiber length parameters using Baer sorter - Span length, 2.5% and 50% span length, uniformity ratio - Estimation of fiber length using Digital fibro graph. Importance of fiber fineness - Airflow principle - Determination of fiber fineness by ATIRA fiber fineness tester. Importance of fiber strength - Strength terminologies-Load, Breaking load, Stress, Mass stress, Tenacity, Strain, Young's modulus, Elongation, Breaking extension - Estimation of bundle fiber strength by Stelometer (CRL Principle). Importance of fiber maturity - List the different methods of determination of fiber maturity - Estimation of fiber maturity by Caustic soda swelling method. Cleaning efficiency - Lint - Trash - invisible loss - Estimation of trash content in cotton by Shirley trash analyzer. Features and working of Advanced Fiber Information System [AFIS] and High Volume Instrument [HVI]. Recognize the Yarn testing methods for determining count, strength, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand advanced fibre testing systems 2 understanding list the important properties of fibre - importance of fiber length - estimation of fiber length parameters using baer sorter - span length, 2.5% and 50% span length, uniformity ratio - estimation of fiber length using digital fibro graph. importance of fiber fineness - airflow principle - determination of fiber fineness by atira fiber fineness tester. importance of fiber strength - strength terminologies-load, breaking load, stress, mass stress, tenacity, strain, young's modulus, elongation, breaking extension - estimation of bundle fiber strength by stelometer (crl principle). importance of fiber maturity - list the different methods of determination of fiber maturity - estimation of fiber maturity by caustic soda swelling method. cleaning efficiency - lint - trash - invisible loss - estimation of trash content in cotton by shirley trash analyzer. features and working of advanced fiber information system [afis] and high volume instrument [hvi]. recognize the yarn testing methods for determining count, strength, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand advanced fibre testing systems 2 Understanding List the important properties of fibre - Importance of fiber length - Estimation of fiber length parameters using Baer sorter - Span length, 2.5% and 50% span length, uniformity ratio - Estimation of fiber length using Digital fibro graph. Importance of fiber fineness - Airflow principle - Determination of fiber fineness by ATIRA

Aspect	What to prepare
	fiber fineness tester. Importance of fiber strength - Strength terminologies-Load, Breaking load, Stress, Mass stress, Tenacity, Strain, Young’s modulus, Elongation, Breaking extension - Estimation of bundle fiber strength by Stelometer (CRL Principle). Importance of fiber maturity - List the different methods of determination of fiber maturity - Estimation of fiber maturity by Caustic soda swelling method. Cleaning efficiency - Lint - Trash - invisible loss - Estimation of trash content in cotton by Shirley trash analyzer. Features and working of Advanced Fiber Information System [AFIS] and High Volume Instrument [HVI]. Recognize the Yarn testing methods for determining count, strength, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Understand advanced fibre testing systems 2 Understanding List the important properties of fibre - Importance of fiber length - Estimation of fiber length parameters using Baer sorter - Span length, 2.5% and 50% span length, uniformity ratio - Estimation of fiber length using Digital fibro graph. Importance of fiber fineness - Airflow principle - Determination of fiber fineness by ATIRA fiber fineness tester. Importance of fiber strength - Strength terminologies-Load, Breaking load, Stress, Mass stress, Tenacity, Strain, Young’s modulus, Elongation, Breaking extension - Estimation of bundle fiber strength by Stelometer (CRL Principle). Importance of fiber maturity - List the different methods of determination of fiber maturity - Estimation of fiber maturity by Caustic soda swelling method. Cleaning efficiency - Lint - Trash - invisible loss - Estimation of trash content in cotton by Shirley trash analyzer. Features and working of Advanced Fiber Information System [AFIS] and High Volume Instrument [HVI]. Recognize the Yarn testing methods for determining count, strength, □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
maturity and trash content in cotton.
             Understand the importance of fibre length,
  M2.01                                                          4
Understanding
             fineness and its determination methods
             Understand the importance and estimation of
                                                                 4
Understanding
  M2.02      fibre strength and maturity.
             Understand the method of estimation of trash
  M2.03                                                             4
Understanding
             content in cotton by Shirley trash analyzer.

  M2.04      Understand advanced fibre testing systems              2
Understanding

             Series Test - I                                        1

Contents:
List the important properties of fibre - Importance of fiber length - Estimation
of fiber
length parameters using Baer sorter - Span length, 2.5% and 50% span length,
uniformity
ratio - Estimation of fiber length using Digital fibro graph. Importance of fiber
fineness -
Airflow principle - Determination of fiber fineness by ATIRA fiber fineness
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

twist, evenness and hairiness of yarn.

This module is presented from the official syllabus scope for Textile Testing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: twist, evenness and hairiness of yarn.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

5 Understanding Yarn count and strength Understand the procedure for yarn twist

5 Understanding Yarn count and strength Understand the procedure for yarn twist is an official topic in Module 3 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding Yarn count and strength Understand the procedure for yarn twist using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding yarn count and strength understand the procedure for yarn twist should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding Yarn count and strength Understand the procedure for yarn twist means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 5 Understanding Yarn count and strength Understand the procedure for yarn twist □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding determination Know the concept of yarn evenness and its

4 Understanding determination Know the concept of yarn evenness and its is an official topic in Module 3 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding determination Know the concept of yarn evenness and its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding determination know the concept of yarn evenness and its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding determination Know the concept of yarn evenness and its means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-[illegible] 4 Understanding determination Know the concept of yarn evenness and its [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding method of determination Understand the concept of testing yarn

4 Understanding method of determination Understand the concept of testing yarn is an official topic in Module 3 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding method of determination Understand the concept of testing yarn using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding method of determination understand the concept of testing yarn should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding method of determination Understand the concept of testing yarn means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding method of determination Understand the concept of testing yarn □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Understanding hairiness. Yarn count - Determination of yarn count by Beesley ’s balance-Importance of yarn strength - factors affecting yarn strength - Estimation of single yarn strength by Uster Tensorapid (CRE Principle) - CSP - Working of lea strength tester (CRT Principle) Importance of Yarn twist - Definition of twist, S twist, Z twist, twist on twist, weft on twist - Different methods of twist determination - Estimation of twist in yarn by tension type twist tester and take-up twist tester. Importance of evenness of the yarn - terms associated with evenness - random variation, periodic variation - Method of determination of yarn evenness - working principle of Uster evenness tester (UT5) - Parameters for expressing unevenness - U%, CV% and Imperfections. Classification of basic classimat faults - Yarn hairiness - Determination of yarn hairiness by Zweigle G 565 Hairiness tester - Comparison between Zweigle S3 value and UT3 hairiness index value. Discuss the Fabric testing methods for determining various fabric

3 Understanding hairiness. Yarn count - Determination of yarn count by Beesley ’s balance-Importance of yarn strength - factors affecting yarn strength - Estimation of single yarn strength by Uster Tensorapid (CRE Principle) - CSP - Working of lea strength tester (CRT Principle) Importance of Yarn twist - Definition of twist, S twist, Z twist, twist on twist, weft on twist - Different methods of twist determination - Estimation of twist in yarn by tension type twist tester and take-up twist tester. Importance of evenness of the yarn - terms associated with evenness - random variation, periodic variation - Method of determination of yarn evenness - working principle of Uster evenness tester (UT5) - Parameters for expressing unevenness - U%, CV% and Imperfections. Classification of basic classimat faults - Yarn hairiness - Determination of yarn hairiness by Zweigle G 565 Hairiness tester - Comparison between Zweigle S3 value and UT3 hairiness index value. Discuss the Fabric testing methods for determining various fabric is an official topic in Module 3 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding hairiness. Yarn count - Determination of yarn count by Beesley ’s balance-Importance of yarn strength - factors affecting yarn strength - Estimation of single yarn strength by Uster Tensorapid (CRE Principle) - CSP - Working of lea strength tester (CRT Principle) Importance of Yarn twist - Definition of twist, S twist, Z twist, twist on twist, weft on twist - Different methods of twist determination - Estimation of twist in yarn by tension type twist tester and take-up twist tester. Importance of evenness of the yarn - terms associated with evenness - random variation, periodic variation - Method of determination of yarn evenness - working principle of Uster evenness tester (UT5) - Parameters for expressing unevenness - U%, CV% and Imperfections. Classification of basic classimat faults - Yarn hairiness - Determination of yarn hairiness by Zweigle G 565 Hairiness tester - Comparison between Zweigle S3 value and UT3 hairiness index value. Discuss the Fabric testing methods for determining various fabric using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding hairiness. yarn count - determination of yarn count by beesley ’s balance-importance of yarn strength - factors affecting yarn strength - estimation of single yarn strength by uster tensorapid (cre principle) - csp - working of lea strength tester (crt principle) importance of yarn twist - definition of twist, s twist, z twist, twist on twist, weft on twist - different methods of twist determination - estimation of twist in yarn by tension type twist tester and take-up twist tester. importance of evenness of the yarn - terms associated with evenness - random variation, periodic variation - method of determination of yarn evenness - working principle of uster evenness tester (ut5) - parameters for expressing unevenness - u%, cv% and imperfections. classification of basic classimat faults - yarn hairiness - determination of yarn hairiness by zweigle g 565 hairiness tester - comparison between zweigle s3 value and ut3 hairiness index value. discuss the fabric testing methods for determining various fabric should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding hairiness. Yarn count - Determination of yarn count by Beesley ’s balance-Importance of yarn strength - factors affecting yarn strength - Estimation of single yarn strength by Uster Tensorapid (CRE Principle) - CSP - Working of lea strength tester (CRT Principle) Importance of Yarn twist - Definition of twist, S twist, Z twist, twist on twist, weft on twist - Different methods of twist determination - Estimation of twist in yarn by tension type twist tester and take-up twist tester. Importance of evenness of the yarn - terms associated with evenness -

Aspect	What to prepare
	random variation, periodic variation - Method of determination of yarn evenness - working principle of Uster evenness tester (UT5) - Parameters for expressing unevenness - U%, CV% and Imperfections. Classification of basic classimat faults - Yarn hairiness - Determination of yarn hairiness by Zweigle G 565 Hairiness tester - Comparison between Zweigle S3 value and UT3 hairiness index value. Discuss the Fabric testing methods for determining various fabric means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding hairiness. Yarn count - Determination of yarn count by Beesley ’s balance-Importance of yarn strength - factors affecting yarn strength - Estimation of single yarn strength by Uster Tensorapid (CRE Principle) - CSP - Working of lea strength tester (CRT Principle) Importance of Yarn twist - Definition of twist, S twist, Z twist, twist on twist, weft on twist - Different methods of twist determination - Estimation of twist in yarn by tension type twist tester and take-up twist tester. Importance of evenness of the yarn - terms associated with evenness - random variation, periodic variation - Method of determination of yarn evenness - working principle of Uster evenness tester (UT5) - Parameters for expressing unevenness - U%, CV% and Imperfections. Classification of basic classimat faults - Yarn hairiness - Determination of yarn hairiness by Zweigle G 565 Hairiness tester - Comparison between Zweigle S3 value and UT3 hairiness index value. Discuss the Fabric testing methods for determining various fabric □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
twist, evenness and hairiness of yarn.
             Understand the methods of determining
  M3.01                                                             5
Understanding
             Yarn count and strength
             Understand the procedure for yarn twist
  M3.02                                                             4
Understanding
             determination
             Know the concept of yarn evenness and its
  M3.03                                                             4
Understanding
             method of determination
             Understand the concept of testing yarn
  M3.04                                                             3
Understanding
             hairiness.

Contents:
Yarn count - Determination of yarn count by Beesley ’s balance-Importance of yarn
strength - factors affecting yarn strength - Estimation of single yarn strength
by Uster
Tensorapid (CRE Principle) - CSP - Working of lea strength tester (CRT Principle)
Importance of Yarn twist - Definition of twist, S twist, Z twist, twist on twist,
weft on twist
- Different methods of twist determination - Estimation of twist in yarn by
tension type
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

properties.

This module is presented from the official syllabus scope for Textile Testing. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: properties.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Remembering characteristics 3 Understand the methods of determining

Remembering characteristics 3 Understand the methods of determining is an official topic in Module 4 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Remembering characteristics 3 Understand the methods of determining using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, remembering characteristics 3 understand the methods of determining should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Remembering characteristics 3 Understand the methods of determining means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ Remembering characteristics 3 Understand the methods of determining □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Fabric dimension, threads density, fabric Understanding 4 weight, cover factor and crimp. Understand the methods of determining

Fabric dimension, threads density, fabric Understanding 4 weight, cover factor and crimp. Understand the methods of determining is an official topic in Module 4 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Fabric dimension, threads density, fabric Understanding 4 weight, cover factor and crimp. Understand the methods of determining using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, fabric dimension, threads density, fabric understanding 4 weight, cover factor and crimp. understand the methods of determining should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Fabric dimension, threads density, fabric Understanding 4 weight, cover factor and crimp. Understand the methods of determining means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Fabric dimension, threads density, fabric Understanding 4 weight, cover factor and crimp. Understand the methods of determining □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Fabric strength, abrasion resistance and Understanding 4 stiffness. Understand the methods of determining

Fabric strength, abrasion resistance and Understanding 4 stiffness. Understand the methods of determining is an official topic in Module 4 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Fabric strength, abrasion resistance and Understanding 4 stiffness. Understand the methods of determining using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, fabric strength, abrasion resistance and understanding 4 stiffness. understand the methods of determining should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Fabric strength, abrasion resistance and Understanding 4 stiffness. Understand the methods of determining means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Fabric strength, abrasion resistance and Understanding 4 stiffness. Understand the methods of determining □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

drape co-efficient, crease property and air Understanding 3 permeability of fabric Fabric properties and characteristics - Dimensional properties of Fabric – fabric thickness - Threads density - Fabric weight - Count of the yarn used for fabric manufacture - Determination of cover factor - Warp cover factor, Weft cover factor and Cloth cover factor - Crimp in yarn - Crimp percentage and crimp amplitude - Shirley yarn crimp tester - Define pill and causes of pill formation - Fabric strength - Determination of Fabric tensile strength by CRT principle - Elmendorf Tearing strength tester - Hydraulic bursting strength tester - Fabric abrasion resistance - Types of abrasions - Martindale abrasion tester - Fabric stiffness - Bending length, flexural rigidity and bending modulus - Cantilever principle - Shirley stiffness tester - Drape and drape co-efficient -Determination of drape by Drape meter - Crease resistance and Crease recovery - Shirley crease recovery tester - Define air and water permeability - air permeability, air resistance and air porosity - Shirley air permeability tester Text / Reference: T/R Book Title/Author T1 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition R2 Arindam Basu - Textile Testing Fiber, Yarn & Fabric, SITRA Publication 2019 R3 Butterworth & Grover - Hand book of Quality control and Textile testing R4 ISO - Hand book of textile testing R5 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition Online resources: Sl.No Website Link 1 www.textilelearner.blogspot.com 2 http://mytextilenotes.blogspot.in 3 http://nptel.ac.in/ 4 https://www.astm.org/Standards/textile-standards.html

drape co-efficient, crease property and air Understanding 3 permeability of fabric Fabric properties and characteristics - Dimensional properties of Fabric – fabric thickness - Threads density - Fabric weight - Count of the yarn used for fabric manufacture - Determination of cover factor - Warp cover factor, Weft cover factor and Cloth cover factor - Crimp in yarn - Crimp percentage and crimp amplitude - Shirley yarn crimp tester - Define pill and causes of pill formation - Fabric strength - Determination of Fabric tensile strength by CRT principle - Elmendorf Tearing strength tester - Hydraulic bursting strength tester - Fabric abrasion resistance - Types of abrasions - Martindale abrasion tester - Fabric stiffness - Bending length, flexural rigidity and bending modulus - Cantilever principle - Shirley stiffness tester - Drape and drape co-efficient -Determination of drape by Drape meter - Crease resistance and Crease recovery - Shirley crease recovery tester - Define air and water permeability - air permeability, air resistance and air porosity - Shirley air permeability tester Text / Reference: T/R Book Title/Author T1 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition R2 Arindam Basu - Textile Testing Fiber, Yarn & Fabric, SITRA Publication 2019 R3 Butterworth & Grover - Hand book of Quality control and Textile testing R4 ISO - Hand book of textile testing R5 J.E. BOOTH - Principles of Textile Testing-

Butterworth third edition Online resources: Sl.No Website Link 1 www.textilelearner.blogspot.com 2 http://mytextilenotes.blogspot.in 3 http://nptel.ac.in/ 4 https://www.astm.org/Standards/textile-standards.html is an official topic in Module 4 of Textile Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify drape co-efficient, crease property and air Understanding 3 permeability of fabric Fabric properties and characteristics - Dimensional properties of Fabric – fabric thickness - Threads density - Fabric weight - Count of the yarn used for fabric manufacture - Determination of cover factor - Warp cover factor, Weft cover factor and Cloth cover factor - Crimp in yarn - Crimp percentage and crimp amplitude - Shirley yarn crimp tester - Define pill and causes of pill formation - Fabric strength - Determination of Fabric tensile strength by CRT principle - Elmendorf Tearing strength tester - Hydraulic bursting strength tester - Fabric abrasion resistance - Types of abrasions - Martindale abrasion tester - Fabric stiffness - Bending length, flexural rigidity and bending modulus - Cantilever principle - Shirley stiffness tester - Drape and drape co-efficient -Determination of drape by Drape meter - Crease resistance and Crease recovery - Shirley crease recovery tester - Define air and water permeability - air permeability, air resistance and air porosity - Shirley air permeability tester Text / Reference: T/R Book Title/Author T1 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition R2 Arindam Basu - Textile Testing Fiber, Yarn & Fabric, SITRA Publication 2019 R3 Butterworth & Grover - Hand book of Quality control and Textile testing R4 ISO - Hand book of textile testing R5 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition Online resources: Sl.No Website Link 1 www.textilelearner.blogspot.com 2 http://mytextilenotes.blogspot.in 3 http://nptel.ac.in/ 4 https://www.astm.org/Standards/textile-standards.html using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, drape co-efficient, crease property and air understanding 3 permeability of fabric fabric properties and characteristics - dimensional properties of fabric – fabric thickness - threads density - fabric weight - count of the yarn used for fabric manufacture - determination of cover factor - warp cover factor, weft cover factor and cloth cover factor - crimp in yarn - crimp percentage and crimp amplitude - shirley yarn crimp tester - define pill and causes of pill formation - fabric strength - determination of fabric tensile strength by crt principle - elmendorf tearing strength tester - hydraulic bursting strength tester - fabric abrasion resistance - types of abrasions - martindale abrasion tester - fabric stiffness - bending length, flexural rigidity and bending modulus - cantilever principle - shirley stiffness tester - drape and drape co-efficient -determination of drape by drape meter - crease resistance and crease recovery - shirley crease recovery tester - define air and water permeability - air permeability, air resistance and air porosity - shirley air permeability tester text / reference: t/r book title/author t1 j.e. booth - principles of textile testing- butterworth third edition r2 arindam basu - textile testing fiber, yarn & fabric, sitra publication 2019 r3 butterworth & grover - hand book of quality control and textile testing r4 iso - hand book of textile testing r5 j.e. booth - principles of textile testing- butterworth third edition online resources: sl.no website link 1 www.textilelearner.blogspot.com 2 http://mytextilenotes.blogspot.in 3 http://nptel.ac.in/ 4 https://www.astm.org/standards/textile-standards.html should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What drape co-efficient, crease property and air Understanding 3 permeability of fabric Fabric properties and characteristics - Dimensional properties of Fabric – fabric thickness - Threads density - Fabric weight - Count of the yarn used for fabric manufacture - Determination of cover factor - Warp cover factor, Weft cover factor and Cloth cover factor - Crimp in yarn - Crimp percentage and crimp amplitude - Shirley yarn crimp tester - Define pill and causes of pill formation - Fabric strength - Determination of Fabric tensile strength by CRT principle - Elmendorf Tearing strength tester - Hydraulic bursting strength tester - Fabric abrasion resistance - Types of abrasions - Martindale abrasion tester - Fabric stiffness - Bending length, flexural rigidity and bending modulus - Cantilever principle - Shirley stiffness tester - Drape and drape co-efficient - Determination of drape by Drape meter - Crease resistance and Crease recovery - Shirley crease recovery tester - Define air and water permeability - air permeability, air resistance and air porosity - Shirley air permeability tester Text / Reference: T/R Book Title/Author T1 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition R2 Arindam Basu - Textile Testing Fiber, Yarn & Fabric, SITRA Publication 2019 R3 Butterworth & Grover - Hand book of Quality control and Textile testing R4 ISO - Hand book of textile testing R5 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition Online resources: Sl.No Website Link 1 www.textilelearner.blogspot.com 2 http://mytextilenotes.blogspot.in 3 http://nptel.ac.in/ 4 https://www.astm.org/Standards/textile-standards.html means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ drape co-efficient, crease property and air Understanding 3 permeability of fabric Fabric properties and characteristics - Dimensional properties of Fabric – fabric thickness - Threads density - Fabric weight - Count of the yarn used for fabric manufacture - Determination of cover factor - Warp cover factor, Weft cover factor and Cloth cover factor - Crimp in yarn - Crimp percentage and crimp amplitude - Shirley yarn crimp tester - Define pill and causes of pill formation - Fabric strength - Determination of Fabric tensile strength by CRT principle - Elmendorf Tearing strength tester - Hydraulic bursting strength tester - Fabric abrasion resistance - Types of abrasions - Martindale abrasion tester - Fabric stiffness - Bending length, flexural rigidity and bending modulus - Cantilever principle - Shirley stiffness tester - Drape and drape co-efficient -Determination of drape by Drape meter - Crease resistance and Crease recovery - Shirley crease recovery tester - Define air and water permeability - air permeability, air resistance and air porosity - Shirley air permeability tester Text / Reference: T/R Book Title/Author T1 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition R2 Arindam Basu - Textile Testing Fiber, Yarn & Fabric, SITRA Publication 2019 R3 Butterworth & Grover - Hand book of Quality control and Textile testing R4 ISO - Hand book of textile testing R5 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition Online resources: Sl.No Website Link 1 www.textilelearner.blogspot.com 2 http://mytextilenotes.blogspot.in 3 http://nptel.ac.in/ 4 https://www.astm.org/Standards/textile-standards.html □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
properties.
             Understand the different fabric properties and
  M4.01
Remembering
             characteristics                                      3
             Understand the methods of determining
  M4.02      Fabric dimension, threads density, fabric
Understanding
                                                                  4
             weight, cover factor and crimp.
             Understand the methods of determining
  M4.03      Fabric strength, abrasion resistance and
Understanding
                                                                  4
             stiffness.
             Understand the methods of determining
  M4.04      drape co-efficient, crease property and air
Understanding
                                                                  3
             permeability of fabric
             Series Test - II                                     1

Contents:
Fabric properties and characteristics - Dimensional properties of Fabric – fabric
thickness
- Threads density - Fabric weight - Count of the yarn used for fabric manufacture
-
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Understand the objectives of Textile Testing 2 Remembering Understand the elements of statistics and its. (3 marks)

Answer: Begin with the standard definition or identity of *Understand the objectives of Textile Testing 2 Remembering Understand the elements of statistics and its*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Applying application in textile testing. Understand the influence of moisture and. (3 marks)

Answer: Begin with the standard definition or identity of *6 Applying application in textile testing. Understand the influence of moisture and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding Humidity on textile materials. Understand the measurement of Humidity, Understanding. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Humidity on textile materials. Understand the measurement of Humidity, Understanding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Moisture content and regain Objectives of Textile Testing - Define statistics - Selection of Sample for testing - population and sample in statistics - frequency distribution - Graphical representation of frequency distribution - Histogram, frequency polygon and frequency curve - Measures of central Tendency - mean, median and mode - Measures of Dispersion - Range, standard deviation, CV%, Variance . Estimation of Measures of central Tendency & Dispersion. Humidity - importance of humidity in textile testing - expressions of humidity - Absolute humidity and Relative humidity - Measurement of Humidity of the atmosphere by wet and dry bulb and sling hygrometer - standard testing atmosphere - Estimation of Moisture content and regain of textile materials by conditioning oven and electronic moisture

meter - Standard moisture regain of important textile fibers - Estimation of standard regain of blends P/C, P/W, P/V - Effect of moisture regain on fibre properties. Explain the procedure for testing Fiber length, strength, fineness,. (3 marks)

Answer: Begin with the standard definition or identity of *4 Moisture content and regain Objectives of Textile Testing - Define statistics - Selection of Sample for testing - population and sample in statistics - frequency distribution - Graphical representation of frequency distribution - Histogram, frequency polygon and frequency curve - Measures of central Tendency - mean, median and mode - Measures of Dispersion - Range, standard deviation, CV%, Variance . Estimation of Measures of central Tendency & Dispersion. Humidity - importance of humidity in textile testing - expressions of humidity - Absolute humidity and Relative humidity - Measurement of Humidity of the atmosphere by wet and dry bulb and sling hygrometer - standard testing atmosphere - Estimation of Moisture content and regain of textile materials by conditioning oven and electronic moisture meter - Standard moisture regain of important textile fibers - Estimation of standard regain of blends P/C, P/W, P/V - Effect of moisture regain on fibre properties. Explain the procedure for testing Fiber length, strength, fineness,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 4 Understanding fineness and its determination methods Understand the importance and estimation of. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding fineness and its determination methods Understand the importance and estimation of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain fibre strength and maturity. Understand the method of estimation of trash. (3 marks)

Answer: Begin with the standard definition or identity of *fibre strength and maturity. Understand the method of estimation of trash*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding content in cotton by Shirley trash analyzer.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding content in cotton by Shirley trash analyzer.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Understand advanced fibre testing systems 2 Understanding List the important properties of fibre - Importance of fiber length - Estimation of fiber length parameters using Baer sorter - Span length, 2.5% and 50% span length, uniformity ratio - Estimation of fiber length using Digital fibro graph. Importance of fiber fineness - Airflow principle - Determination of fiber fineness by ATIRA fiber fineness tester. Importance of fiber strength - Strength terminologies-Load, Breaking load, Stress, Mass stress, Tenacity, Strain, Young’s modulus, Elongation, Breaking extension - Estimation of bundle fiber strength by Stelometer (CRL Principle). Importance of fiber maturity - List the different methods of determination of fiber maturity - Estimation of fiber maturity by Caustic soda swelling method. Cleaning efficiency - Lint - Trash - invisible loss - Estimation of trash content in cotton by Shirley trash analyzer. Features and working of Advanced Fiber Information System [AFIS] and High Volume Instrument [HVI]. Recognize the Yarn testing methods for determining count, strength,. (3 marks)

Answer: Begin with the standard definition or identity of *Understand advanced fibre testing systems 2 Understanding List the important properties of fibre - Importance of fiber length - Estimation of fiber length parameters using Baer sorter - Span length, 2.5% and 50% span length, uniformity ratio - Estimation of fiber length using Digital fibro graph. Importance of fiber fineness - Airflow principle - Determination of fiber fineness by ATIRA fiber fineness tester. Importance of fiber strength - Strength terminologies-Load, Breaking load, Stress, Mass stress, Tenacity, Strain, Young's modulus, Elongation, Breaking extension - Estimation of bundle fiber strength by Stelometer (CRL Principle). Importance of fiber maturity - List the different methods of determination of fiber maturity - Estimation of fiber maturity by Caustic soda swelling method. Cleaning efficiency - Lint - Trash - invisible loss - Estimation of trash content in cotton by Shirley trash analyzer. Features and working of Advanced Fiber Information System [AFIS] and High Volume Instrument [HVI]. Recognize the Yarn testing methods for determining count, strength,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 5 Understanding Yarn count and strength Understand the procedure for yarn twist. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding Yarn count and strength Understand the procedure for yarn twist*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding determination Know the concept of yarn evenness and its. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding determination Know the concept of yarn evenness and its*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding method of determination Understand the concept of testing yarn. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding method of determination Understand the concept of testing yarn*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding hairiness. Yarn count - Determination of yarn count by Beesley ’s balance-Importance of yarn strength - factors affecting yarn strength - Estimation of single yarn strength by Uster Tensorapid (CRE Principle) - CSP - Working of lea strength tester (CRT Principle) Importance of Yarn twist - Definition of twist, S twist, Z twist, twist on twist, weft on twist - Different methods of twist determination - Estimation of twist in yarn by tension type twist tester and take-up twist tester. Importance of evenness of the yarn - terms associated with evenness - random variation, periodic variation - Method of determination of yarn evenness - working principle of Uster evenness tester (UT5) - Parameters for expressing unevenness - U%, CV% and Imperfections. Classification of basic classimat faults - Yarn hairiness - Determination of yarn hairiness by Zweigle G 565 Hairiness tester - Comparison between Zweigle S3 value and UT3 hairiness index value. Discuss the Fabric testing methods for determining various fabric. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding hairiness. Yarn count - Determination of yarn count by Beesley ’s balance-Importance of yarn strength - factors affecting yarn strength - Estimation of single yarn strength by Uster Tensorapid (CRE Principle) - CSP - Working of lea strength tester (CRT Principle) Importance of Yarn twist - Definition of twist, S twist, Z twist, twist on twist, weft on twist - Different methods of twist determination - Estimation of twist in yarn by tension type* *twist tester and take-up twist tester. Importance of evenness of the yarn - terms associated with evenness - random variation, periodic variation - Method of determination of yarn evenness - working principle of Uster evenness tester (UT5) - Parameters for expressing unevenness - U%, CV% and Imperfections. Classification of basic classimat faults - Yarn hairiness - Determination of yarn hairiness by Zweigle G 565 Hairiness tester - Comparison between Zweigle S3 value and UT3 hairiness index value. Discuss the Fabric testing methods for determining various fabric*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Remembering characteristics 3 Understand the methods of determining. (3 marks)

Answer: Begin with the standard definition or identity of *Remembering characteristics 3 Understand the methods of determining*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Fabric dimension, threads density, fabric Understanding 4 weight, cover factor and crimp. Understand the methods of determining. (3 marks)

Answer: Begin with the standard definition or identity of *Fabric dimension, threads density, fabric Understanding 4 weight, cover factor and crimp. Understand the methods of determining*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Fabric strength, abrasion resistance and Understanding 4 stiffness. Understand the methods of determining. (3 marks)

Answer: Begin with the standard definition or identity of *Fabric strength, abrasion resistance and Understanding 4 stiffness. Understand the methods of determining*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain drape co-efficient, crease property and air Understanding 3 permeability of fabric Fabric properties and characteristics - Dimensional properties of Fabric – fabric thickness - Threads density - Fabric weight - Count of the yarn used for fabric manufacture - Determination of cover factor - Warp cover factor, Weft cover factor and Cloth cover factor - Crimp in yarn - Crimp percentage and crimp amplitude - Shirley yarn crimp tester - Define pill and causes of pill formation - Fabric strength - Determination of Fabric tensile strength by CRT principle - Elmendorf Tearing strength tester - Hydraulic bursting strength tester - Fabric abrasion resistance - Types of abrasions - Martindale abrasion tester - Fabric stiffness - Bending length, flexural rigidity and bending modulus - Cantilever principle - Shirley stiffness tester - Drape and drape co-efficient -Determination of drape by Drape meter - Crease resistance and Crease recovery - Shirley crease recovery tester - Define air and water permeability - air permeability, air resistance and air porosity - Shirley air permeability tester Text / Reference: T/R Book Title/Author T1 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition R2 Arindam Basu - Textile Testing Fiber, Yarn & Fabric, SITRA Publication 2019 R3 Butterworth & Grover - Hand book of Quality control and Textile testing R4 ISO - Hand book of textile testing R5 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition Online resources: Sl.No Website Link 1 www.textilelearner.blogspot.com 2 http://mytextilenotes.blogspot.in 3 http://nptel.ac.in/ 4 https://www.astm.org/Standards/textile-standards.html. (3 marks)

Answer: Begin with the standard definition or identity of *drape co-efficient, crease property and air Understanding 3 permeability of fabric Fabric properties and characteristics - Dimensional properties of Fabric – fabric thickness - Threads density - Fabric weight - Count of the yarn used for fabric manufacture - Determination of cover factor - Warp cover factor, Weft cover factor and Cloth cover factor - Crimp in yarn - Crimp percentage and crimp amplitude - Shirley yarn crimp tester - Define pill and causes of pill formation - Fabric strength - Determination of Fabric tensile strength by CRT principle - Elmendorf Tearing strength tester - Hydraulic bursting strength tester - Fabric abrasion resistance - Types of abrasions - Martindale abrasion tester - Fabric stiffness - Bending length, flexural rigidity and bending modulus - Cantilever principle - Shirley stiffness tester - Drape and drape co-efficient -Determination of drape by Drape meter - Crease resistance and Crease recovery - Shirley crease recovery tester - Define air and water permeability - air permeability, air resistance and air porosity - Shirley air permeability tester Text / Reference: T/R Book Title/Author T1 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition R2 Arindam Basu - Textile Testing Fiber, Yarn & Fabric, SITRA Publication 2019 R3 Butterworth & Grover - Hand book of Quality control and Textile testing R4 ISO - Hand book of textile testing R5 J.E. BOOTH - Principles of Textile Testing- Butterworth third edition Online resources: Sl.No Website Link 1 www.textilelearner.blogspot.com 2 http://mytextilenotes.blogspot.in 3 http://nptel.ac.in/ 4 https://www.astm.org/Standards/textile-standards.html*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Understand the objectives of Textile Testing 2 Remembering Understand the elements of statistics and its**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding fineness and its determination methods Understand the importance and estimation of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **5 Understanding Yarn count and strength Understand the procedure for yarn twist**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Remembering characteristics 3 Understand the methods of determining**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link www.textilelearner.blogspot.com http://mytextilenotes.blogspot.in http://nptel.ac.in/
- https://www.astm.org/Standards/textile-standards.html

IN-PAGE SEARCH

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